

TARUN S

Python Developer Intern | Python | FastAPI | REST APIs | Backend

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PROFESSIONAL SUMMARY

Computer Science undergraduate with rigorous Python software engineering fundamentals and hands-on experience designing high-throughput REST APIs, backend architectures, and database services using FastAPI, Django, and PostgreSQL. Adept in data processing pipelines, Docker containerization, and clean OOP principles.

TECHNICAL SKILLS

- ? **Programming:** Python (Strong), Java, JavaScript, C++, SQL
- ? **AI & Machine Learning:** Machine Learning, Supervised Learning, Retrieval-Augmented Generation (RAG), NLP, Vector Embeddings, Semantic Search, Model Evaluation
- ? **Data Science & Libraries:** NumPy, Pandas, Scikit-learn (TF-IDF, Text Classification), Data Preprocessing, Exploratory Data Analysis (EDA)
- ? **ML Frameworks & AI Tools:** Scikit-learn, ChromaDB, OpenAI APIs (Whisper, Embeddings), PyTorch (Foundational), TensorFlow (Foundational)
- ? **Web & Backend:** FastAPI, Django, REST APIs, PostgreSQL, SQLite, Node.js, Angular, React
- ? **Tools & Concepts:** Git, GitHub, Docker, VS Code, Object-Oriented Programming (OOP), Agile Development

AI/ML & TECHNICAL PROJECTS

Spectrum AI — AI-Powered RAG & Document Intelligence Platform

- ? Engineered a Python AI learning platform utilizing OpenAI Whisper for speech-to-text audio transcription and multi-stage text chunking workflows.
- ? Implemented a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB vector database and semantic search embeddings (text-embedding-3-small) for context-aware Q&A.
- ? Developed automated data pipelines integrating speech transcription, vector storage, and LLM APIs to generate structured study notes, interactive quizzes, and dynamic knowledge graph visualizations.

AI Collaboration & Meeting Platform

- ? Built backend RESTful services and relational database schemas in PostgreSQL for real-time collaboration and session management.
- ? Optimized database query performance and API endpoints to ensure seamless data transmission and high application reliability.

HyperTrade AI — Predictive Market Analytics & Decision Engine

- ? Engineered a Python quantitative analytics engine computing technical indicators including RSI, 5/20-day Moving Averages (SMA), and price volatility metrics.
- ? Applied NumPy and Pandas for data preprocessing, time-series data cleaning, and real-time market data evaluation.
- ? Developed a multi-factor algorithmic scoring system, incorporating model training and evaluation, to deliver automated Buy/Hold/Sell recommendations with risk assessment.

EDUCATION

B.Tech in Computer Science and Engineering (3rd Year | Expected Graduation: 2028)

Amrita Vishwa Vidyapeetham | Location: Chennai, India

Relevant Coursework: Machine Learning Foundations, Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems

EXPERIENCE & LEADERSHIP

Product & Support Lead ? Early Stage Tech Startup (CLABROOM) | Chennai, India

- ? Front end developer intern at Quantum AI and electronics labs.
- ? Vocal Bridge: Worked with an early-stage AI automation and voice-agent startup, developing solutions for automated voice interactions and business process automation.
- ? Managed technical feature iterations and customer issue resolution, bridging user feedback with backend engineering tasks.
- ? Collaborated with cross-functional development teams in agile sprint planning to accelerate deployment cycle.